TECHNICAL REPORT

NHS Appointment Utilisation & Missed Appointments

Analysis of NHS appointment data to identify capacity utilisation patterns, missed appointment drivers, and data quality constraints across 106 locations and 7 regions.

Andrew Willacy | August 2025

TOOLS & TECH STACK

Excel	Python	Matplotlib / Seaborn
Data preparation & EDA	Analysis	Charts & dashboards

KEY BUSINESS QUESTIONS

Is capacity adequate? Staff and network vs demand	**How are resources used?** Actual appointment utilisation	**What drives missed appointments?** DNA rates, patterns, and costs

NHS HEALTHCARE: APPOINTMENT, UTILISATION & CAPACITY ANALYSIS

Technical Report - Andrew Willacy - August 2025

Executive Summary

This report presents the findings of an analysis of NHS appointment data, examining capacity utilisation, missed appointment patterns, and resource deployment across primary care in England. Using Python for analysis and Matplotlib and Seaborn for visualisation, the work addressed three core questions: whether adequate capacity has been in place, how resources have actually been utilised, and what is driving missed appointments.

The central finding is not a simple answer to these questions, but a structural constraint on answering them at all: data quality is the primary barrier to understanding NHS performance. Unmapped and unknown values consistently dominate key metrics, the daily capacity benchmark is fundamentally misconfigured, and the Twitter dataset is insufficiently structured for meaningful sentiment analysis in its current form.

Finding	Summary
Scale of dataset	Over 1 million records across 106 Sub-ICB locations and 7 regions; period January 2020 to June 2022
Service composition	General practice accounts for 91.5% of all appointments - national patterns are almost entirely GP-driven
Capacity utilisation	Average 103.2% utilisation across the analysis period; October 2021 the worst month at 120.3% of weekday capacity
Capacity benchmark	Fixed 1.2M daily figure applied uniformly including weekends - fundamentally distorting all utilisation analysis
Missed appointments	4–5% DNA rate consistently; a further 4–5% unmapped; same-day appointments have the highest DNA rate at 19.8%
Regional variation	DNA rates range from 4.7%–5.8% in highest-risk regions; Midlands and London show the highest concentrations
Data quality	Unknown/data quality category dominates key metrics; the primary barrier to reliable insight
Twitter dataset	1,174 records; insufficient structure and volume for meaningful sentiment analysis

Business Problem

The NHS faces increasing demand, constrained resources, and a persistent missed appointment problem. With an ageing population placing growing strain on primary care, understanding whether existing capacity is adequate - and why appointments are missed - is operationally and financially critical.

Each missed appointment carries an estimated cost of £30 and represents a lost care opportunity for another patient. The analysis was commissioned to address two core questions:

- Has there been adequate staff and network capacity?
- What is the actual utilisation of resources?

A secondary objective was to assess whether Twitter (X) data could contribute meaningfully to understanding public sentiment around NHS services, and whether this could inform patient engagement strategies.

Data & Analytical Approach

Datasets

Four datasets were used for the analysis:

Dataset	Key Fields	Purpose
actual_duration.csv	Sub-ICB location, appointment date, actual duration, count of appointments	Duration and volume analysis
appointments_regional.csv	ICB code, appointment month, status, HCP type, mode, time between booking and appointment, count	Regional utilisation, wait times, DNA analysis
national_categories.csv	Appointment date, ICB code, service setting, context type, national category, count	Service type, capacity, seasonal analysis
tweets.csv	Tweet text, hashtags, retweet count, favourite count	Public sentiment (limited applicability)

The national_categories.xlsx file was converted to CSV for consistency before import. The 'appointments_regional' dataset contained 21,604 duplicate records which were removed; no duplicates were found in the other datasets. Cleaned datasets were saved with a _clean suffix. An NHS-ICB names reference file was downloaded from the NHS website and merged with the 'appointments_regional' dataset by ICB ONS code to add readable region names, making location-based analysis more accessible.

Analytical Approach

Python was the primary analytical tool, using Pandas for data manipulation and Matplotlib and Seaborn for visualisation. A GeoJSON boundary file from the Office for National Statistics was used for geographic DNA mapping. Analysis was structured around the three core business questions, extended to cover seasonal patterns, wait time analysis, HCP trends, and geographic variation.

- The 1.2 million daily capacity benchmark was used as provided but identified as a material limitation - applied uniformly across all days including weekends. An adjusted weekday-only capacity analysis was developed to produce a more accurate picture of system pressure.

- Monday appointment counts were noted as potentially inflated due to bank holiday displacement.
- Date formats differed between datasets (dd-mmm-yy in ‘actual_duration’ vs dd/mm/yyyy in ‘national_categories’) - standardised to dd/mm/yyyy across all datasets before analysis.
- The Twitter dataset was reviewed for hashtag frequency patterns only; full sentiment analysis was not feasible given the dataset's structure and volume.
- Datasets were merged where needed for cross-analysis - ‘appointments_regional’ and ‘national_categories’ were joined on ICB ONS code for DNA-by-service-setting analysis.

Key Findings

1. Dataset Overview and Composition

Metric	Finding
Total Sub-ICB locations	106 across 7 regions
ICB locations	42
Highest-volume location	NHS North West London ICB - 12.1 million appointments (4.1% of total)
Dominant service setting	General practice - 91.5% of all appointments
Peak month (volume)	November 2021 - 30.4 million appointments
Lowest month (volume)	August 2021 - 23.9 million appointments
Analysis period	January 2020 - June 2022 (capacity analysis: August 2021 – June 2022)
Twitter records	1,174 records across 10 columns

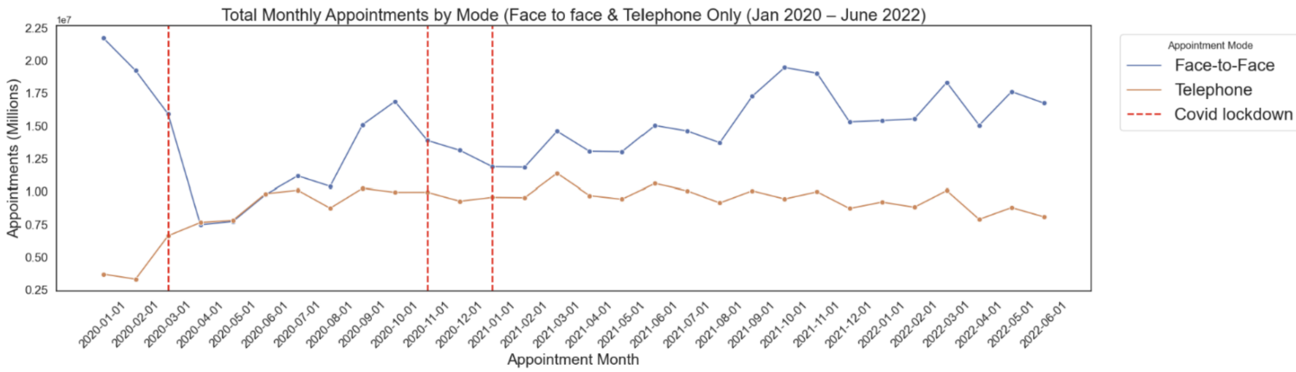
General practice dominates to such an extent that national patterns are almost entirely reflective of GP activity. This has important implications for interpretation: trends, seasonal patterns, and capacity analysis are effectively GP trends. Non-GP service settings are visible only when GP is removed from the analysis.

2. Appointment Modes and HCP Trends

Face-to-face appointments dominated across the full period (figure 1), but the Covid pandemic produced a structural shift in appointment delivery. The March 2020 lockdown caused the steepest single-month drop in face-to-face appointments in the dataset - a dramatic reduction that did not substitute cleanly to telephone. Total appointment volume fell sharply, with telephone appointments increasing but not compensating for the face-to-face loss.

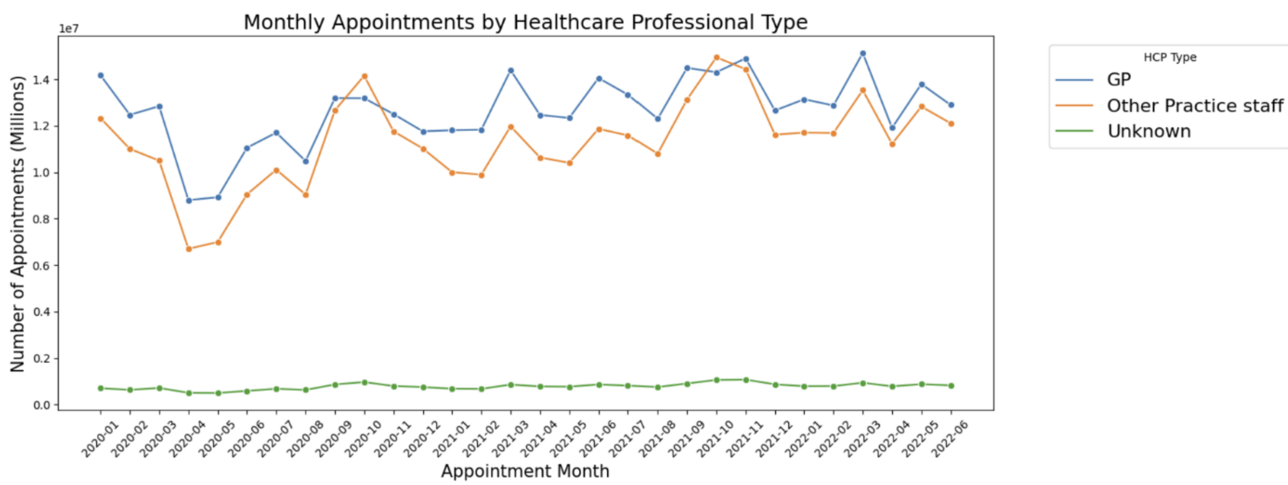
Telephone appointments remained elevated throughout 2021-22, reflecting a lasting change in both patient preference and clinical delivery. By June 2022, telephone and face-to-face appointments were running at substantially different proportions compared to pre-pandemic levels, representing a permanent shift rather than a temporary adjustment.

Figure 1: Monthly Appointments across the time period by Mode



By Healthcare Provider (HCP) type (figure 2), GP appointments accounted for the overwhelming majority of activity. Other Practice Staff and Unknown HCP types showed distinct seasonal patterns - rising in autumn and spring, dipping in winter - in contrast to GP appointment trends where winter pressure was highest. This inverse relationship has implications for capacity planning: different staff types face peak demand at different times of year.

Figure 2: Monthly Appointments across the time period by Healthcare Professional



3. Capacity and Utilisation

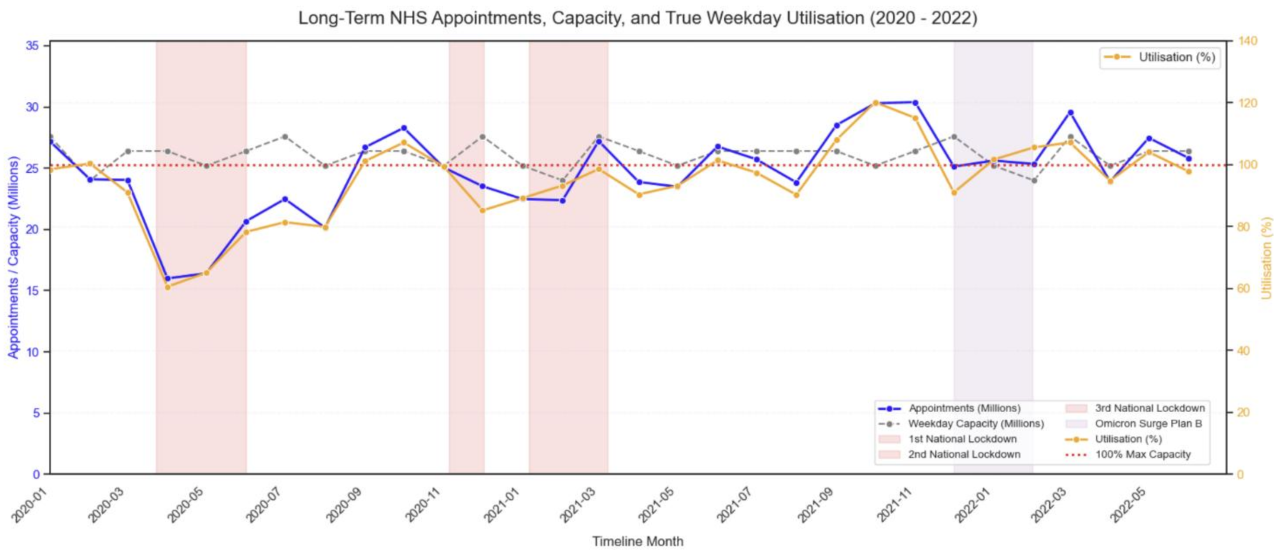
The NHS operates against a stated benchmark of 1.2 million available appointments per day. This benchmark has a fundamental structural problem: it is applied uniformly to every day including weekends, when appointment volumes are a fraction of weekday levels. The effect is to artificially inflate apparent available capacity and materially understate weekday utilisation pressure.

October 2021 was the worst single month: with 21 working days the maximum weekday capacity was 25.2 million appointments, but 30.3 million were delivered - running 20.3% above structural weekday capacity. The system operated at an average of 103.2% of its stated daily capacity across the period, meaning the NHS was consistently exceeding its own benchmark.

Utilisation by Month (August 2021 – June 2022)

Month	Total Appointments	Utilisation vs 1.2M/day benchmark
October 2021	30.3 million	120.3% - highest crisis month (20.3% above weekday structural capacity)
November 2021	30.4 million	115.2%
August 2021	23.9 million	Lowest in period - below benchmark
Average across period	24.7 million	103.2% - system above its own benchmark every month on average

Figure 3: Appointments, Capacity and Utilisation



Day-of-Week Analysis

Tuesday was consistently the highest-volume appointment day, followed by Monday and Wednesday. Monday volumes are partially inflated by bank holiday displacement (figure 6) - appointments rescheduled from bank holidays cluster on the following Monday, distorting the day-of-week picture.

- Capacity was breached on Monday and Tuesday in approximately 90% of weeks across the period
- Wednesday showed marginally better performance
- Friday was the only day where the majority of weeks remained within capacity
- Weekends had near-zero appointments - confirming the 1.2M daily benchmark is meaningless as a uniform daily average

Figure 4: Average Daily Appointments by Day of the week

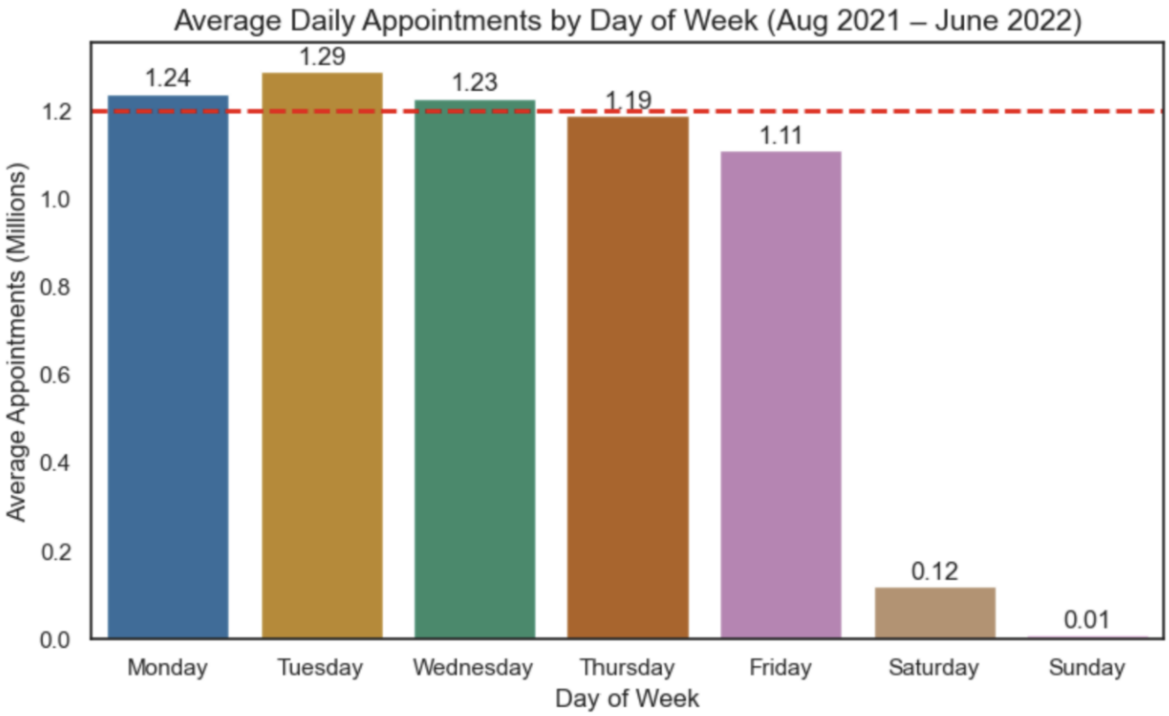


Figure 5: Days when capacity is breached and how often

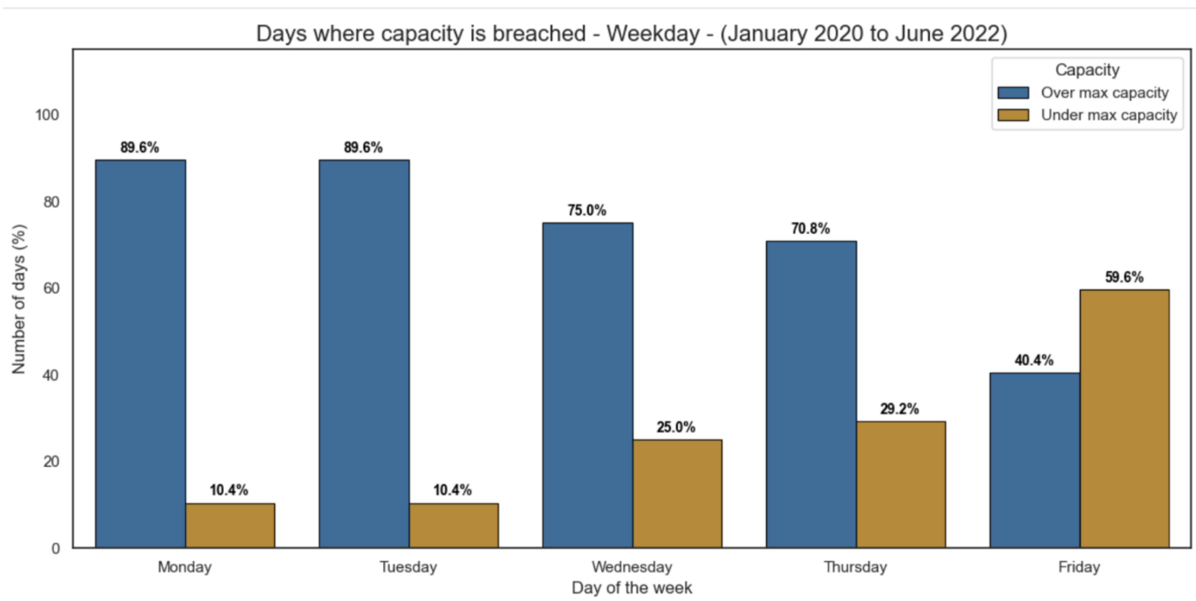
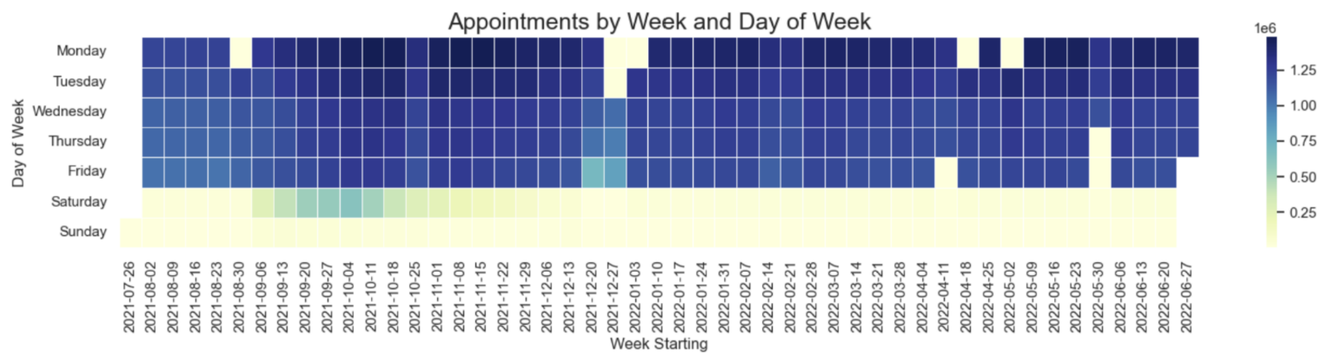
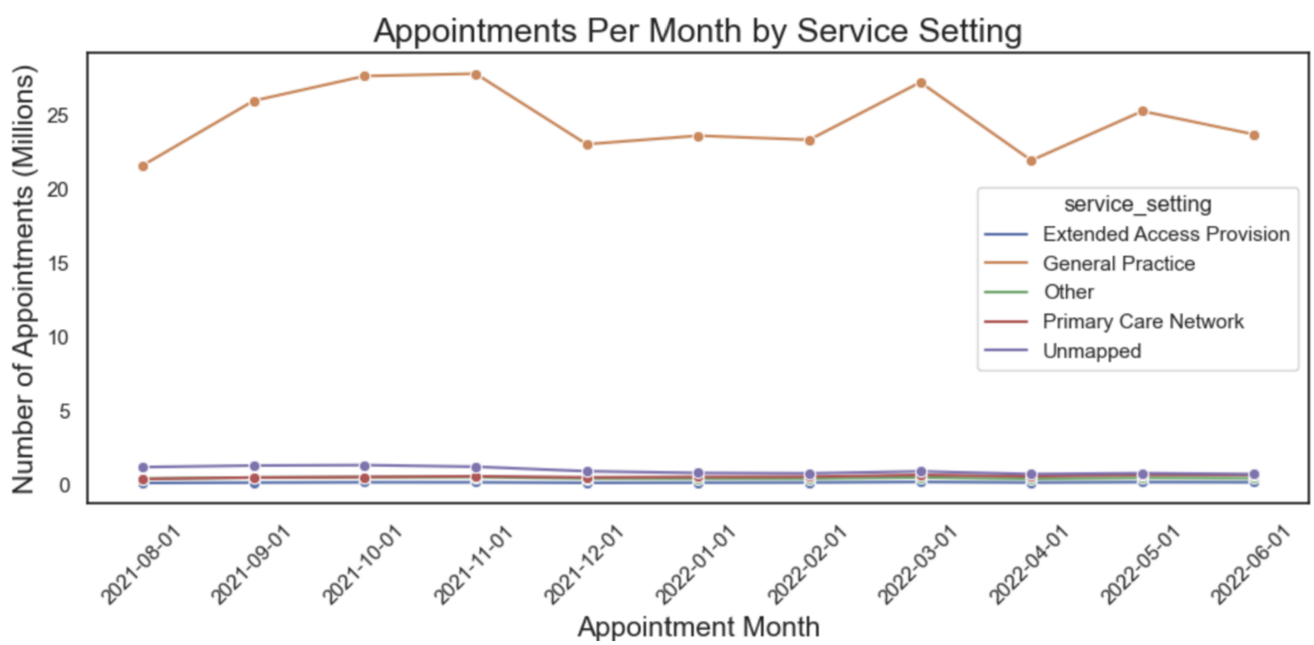


Figure 6: Individual day Appointment levels



Seasonal Patterns

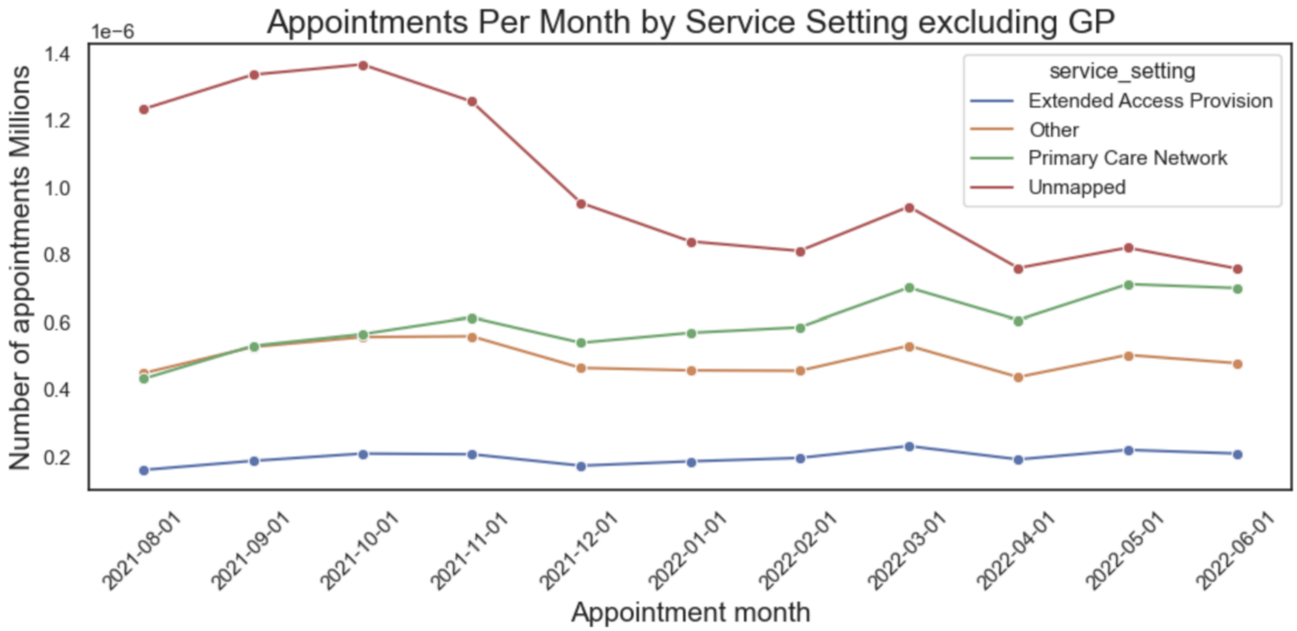
Figure 7: Monthly Appointments by Service Setting



GP appointments dwarf all other service settings making it difficult to extract any useable data from the graph (figure 7). Removing GP appointments allows us to see what is really happening in the other settings (figure 8).

The graph shows how the number of appointments has varied over the period excluding GP. Unmapped dominates, although there is a significant drop during the Autumn of 2021. This could be due to the introduction of the Learn from Patient Safety Events (LFPSE) service, a modernised digital system designed to replace the legacy National Reporting and Learning Service (NRLS) and improve how healthcare organisations collect and learn from safety data.

Figure 8: Monthly Appointments by Service Setting (excluding GP)



Season	Key Findings
Summer (Aug 2021)	Appointment volumes below the 1.2M daily benchmark across all weeks. Clear weekly pattern of declining appointments towards the weekend.
Autumn (Sep–Nov 2021)	Significant increase in GP appointments from October; capacity consistently breached. Predictable seasonal pressure that should be built into capacity planning.
Winter (Dec 2021–Feb 2022)	More variability in appointment numbers. Capacity breached in most weeks but end-of-week numbers lower than other seasons. Christmas week (23 Dec 2021) shows near-zero appointments across all service settings.
Spring (Mar–May 2022)	Appointments concentrated Monday–Wednesday, dropping below capacity in the second half of the week. Early April dip corresponds to Easter bank holidays.

4. Appointment Wait Times

Analysis of the time between booking and attending an appointment revealed same day appointments are the most common, followed a 2 -7 day wait. Waiting more than 28 days is the least common (figure 9).

Figure 9: Appointments by Wait time – Time from booking to attending

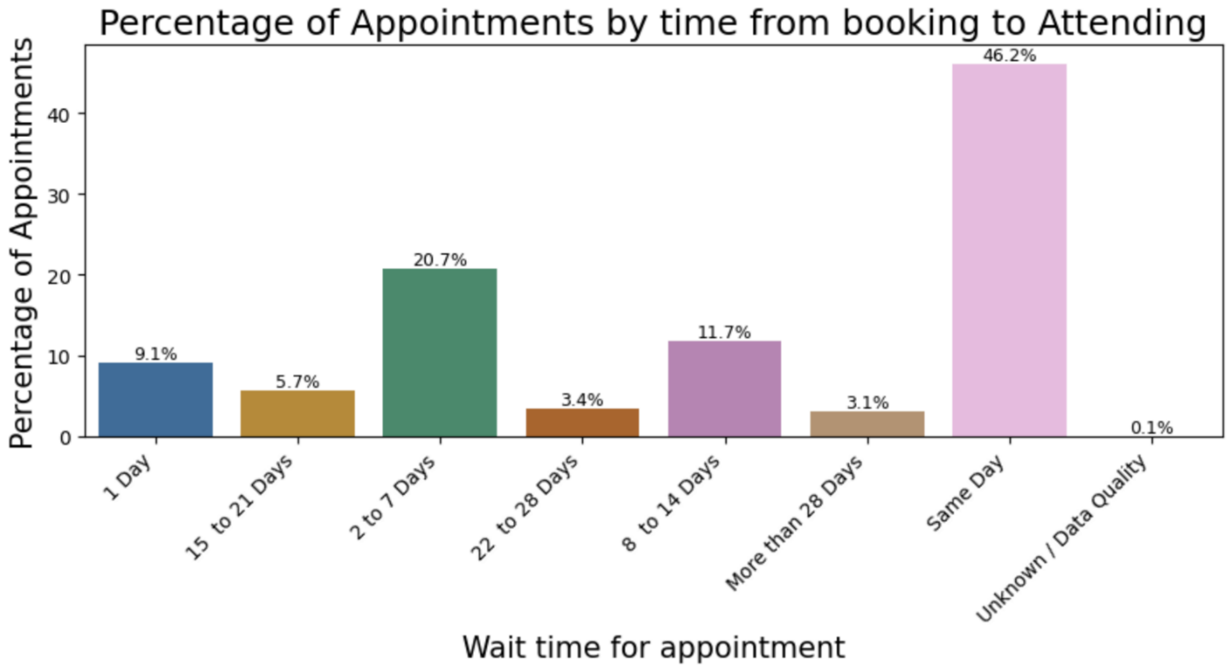


Figure 10: Appointments by Wait time – Variations over the time period

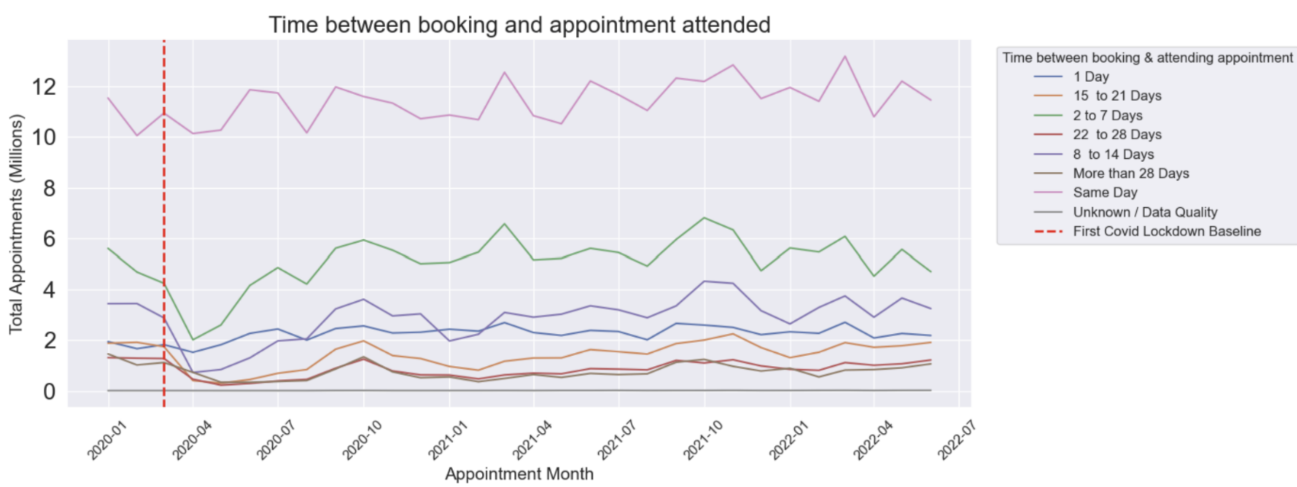
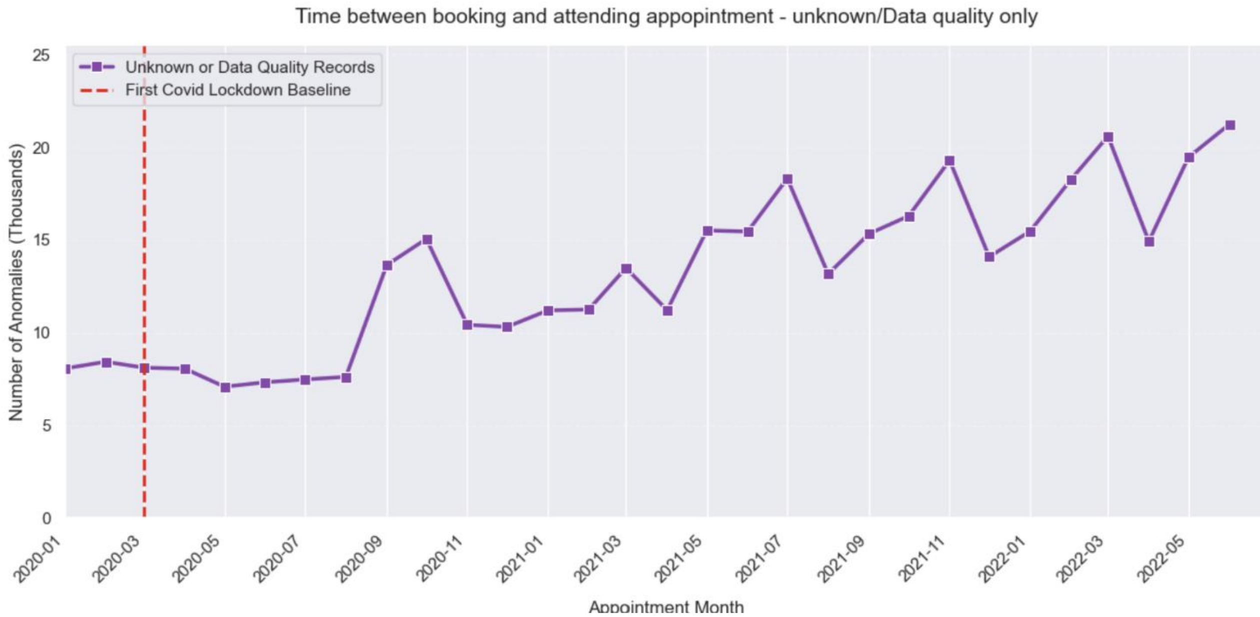


Figure 10 shows how appointment number by wait time varied over the time period. Same day dominates. The ‘Unknown/Data quality’ category appears to be zero or clos to it. It has already been seen that data quality issues are a major problem, so this was investigated further to determine whether this is in fact a true figure. To do this the ‘Unknown/Data quality’ category was extracted and a plot drawn of only this category (figure 11).

Far from being a zero figure the 'Unknown/Data Quality' category for booking-to-appointment times showed a significant spike at the start of the first Covid lockdown in March 2020 - consistent with a systemic recording failure during the disruption period. This category continued to grow across the analysis period, representing an increasing data quality concern for wait time analysis that cannot be resolved through analytical methodology.

Figure 11: Appointments by Wait time – Variations over the time period (‘Unknown/Data quality’ only)



5. Missed Appointments (DNA Analysis)

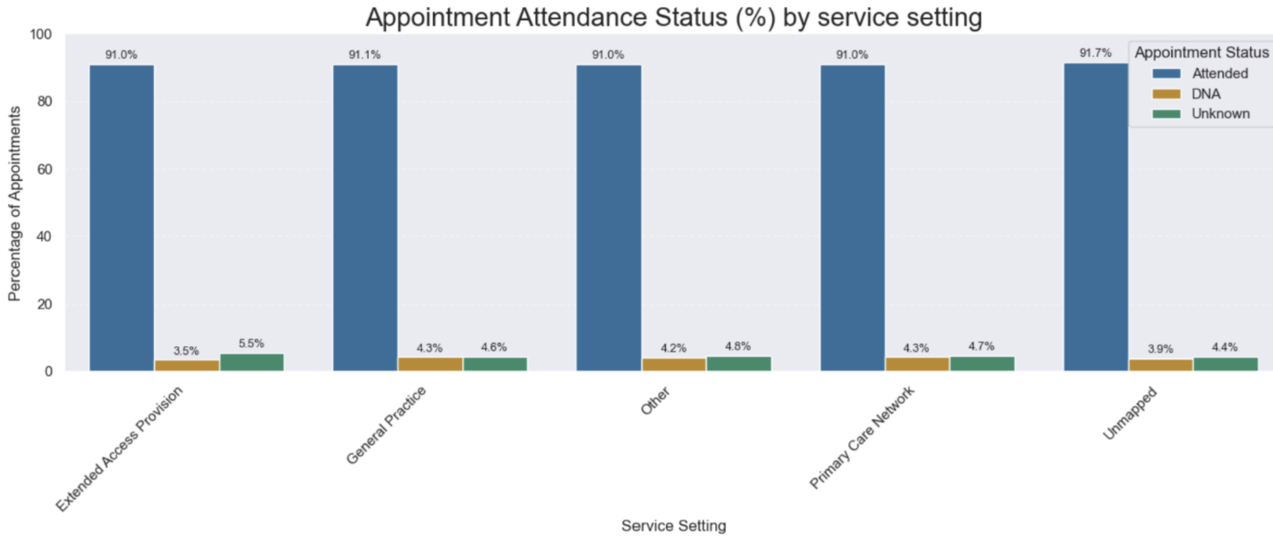
The 4–5% DNA rate is consistent across the analysis period. However, the picture becomes more concerning when the unmapped 'Unknown' category is included - meaning for every confirmed missed appointment there is at least one appointment whose outcome is entirely unknown.

By Service Setting

Metric	Range	Note
DNA rate (all service settings)	3.5%–4.2%	Consistent across all settings
Unknown status rate	4.4%–5.5%	Consistently exceeds the DNA rate - true non-attendance may be significantly higher than the headline figure

GP has the worst DNA rate, but this is to be expected as GP appointment account for over half of all NHS appointments. More concerning is the other service setting DNA rates are not that much lower even though those settings account for far less appointments than GP. Again the ‘Unknown’ status comes into play with it accounting for a higher rate than DNAs across all settings (figure 12).

Figure 12: Appointment Status by Service Setting (Attended, DNA, Unknown)



By Sub-ICB Region

Geographic analysis revealed significant regional variation in DNA rates (figure 13). The top ten highest-DNA regions ranged from 4.7% to 5.8%. Geographic mapping using ONS Sub-ICB boundary data confirmed that the highest DNA rates are concentrated in the Midlands and London - urban areas with more complex demographics, transport challenges, and socioeconomic pressures.

NHS Black Country and NHS Greater Manchester consistently showed the highest DNA rates. This regional variation has direct implications for targeted intervention: reminder systems, rescheduling tools, and public awareness campaigns should be deployed differentially based on regional DNA risk rather than applied uniformly across all areas.

The top 10 Sub-ICB regions by DNA rate also show a consistent pattern in their Unknown status figures, ranging from 3.9%–4.8% - suggesting that data recording quality and non-attendance risk are correlated geographically. This is itself a finding: the regions with the most missed appointments are also the regions where appointment outcomes are least reliably recorded, meaning the true scale of non-attendance in these areas is likely higher than the confirmed figures suggest.

Figure 14 shows the appointment status breakdown for these ten regions in detail. While the absolute DNA percentages appear modest - a 1.1 percentage point spread between 4.7% and 5.8% - at the scale of NHS appointment volumes this represents a material number of lost appointment slots, with significant associated cost. If a conservative £30 per missed appointment is applied and the average NHS region conducts approximately 500,000 appointments per month, a 1% difference in DNA rate equates to approximately 5,000 additional missed appointments per month - or £150,000 in direct cost per region. The geographic concentration of the highest-risk regions therefore makes a strong case for differential resource allocation in intervention design.

Figure 13: DNA % by Sub-ICB Region – Geographic Plot

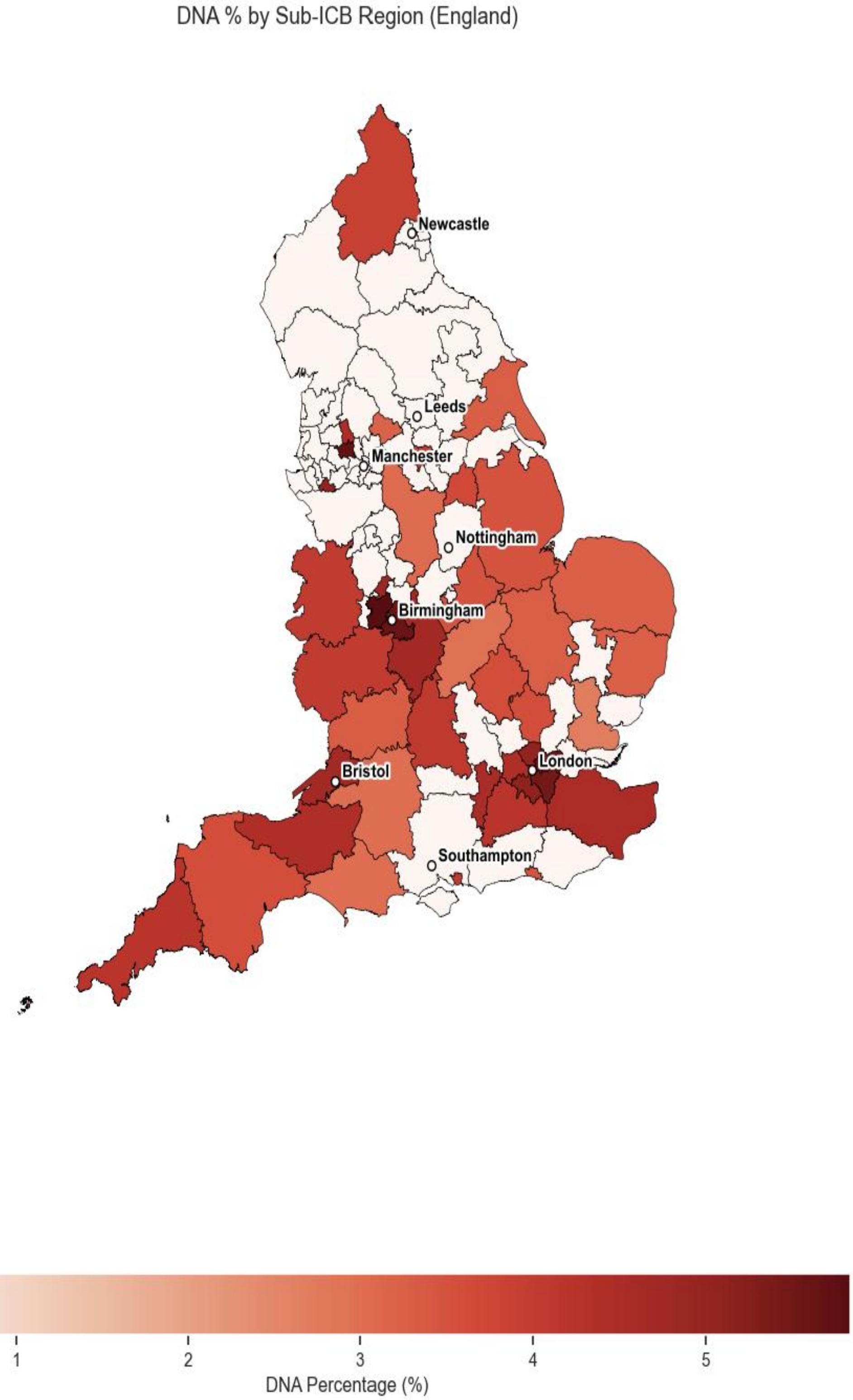
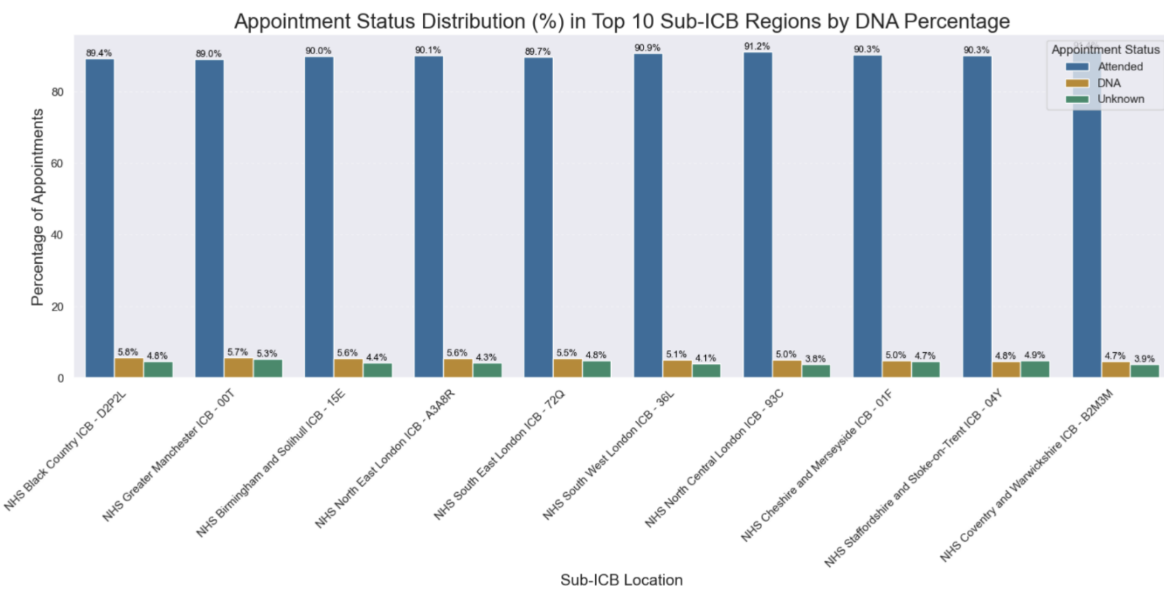


Figure 14: Appointment status by top 10 DNA Sub_ICB Regions



By Wait Time (time between booking and attending an appointment)

This section of the analysis examined the distribution of missed appointments (DNA records only) by wait time category - that is, what proportion of all DNA appointments fell into each booking-to-attendance interval.

The percentage of total DNA appointments by wait time category:

Wait Time Category	% of Total DNAs	Interpretation
2–7 days	Highest	The largest single share of all missed appointments occurred in appointments booked 2–7 days in advance
Same day	~19.8%	Nearly 1 in 5 same-day appointments are not attended. Possible explanations: patient condition changed rapidly after booking, or uncommitted opportunistic bookings. Also raises the question: why are patients not cancelling?
22+ days	Lowest	Appointments booked further in advance generate the smallest share of DNA appointments — patients with longer lead times appear more committed to attending

The pattern across wait time categories suggests a behavioural gradient: patients who book further in advance have more time to reflect on their need for the appointment, to reschedule if circumstances change, and to feel committed to attending. Same-day appointments, by contrast, are often made under urgency- and urgency is transient. A patient who calls for a same-day slot may find their symptoms have eased before the appointment time or may find the practicalities of attending at short notice insurmountable. The absence of a cancellation culture compounds this: if patients don't cancel, the slot is lost with no opportunity for the NHS to reallocate it.

The 2–7-day category generates the most missed appointments in absolute terms, but this partly reflects that it is also the most common booking-to-appointment interval across all appointment types. The same-day DNA rate of ~19.8% is the most operationally significant finding - it represents a genuinely high rate of non-attendance for appointments where there is effectively no advance warning or lead time for the NHS to act.

Figure 15: Missed Appointments by Wait Time

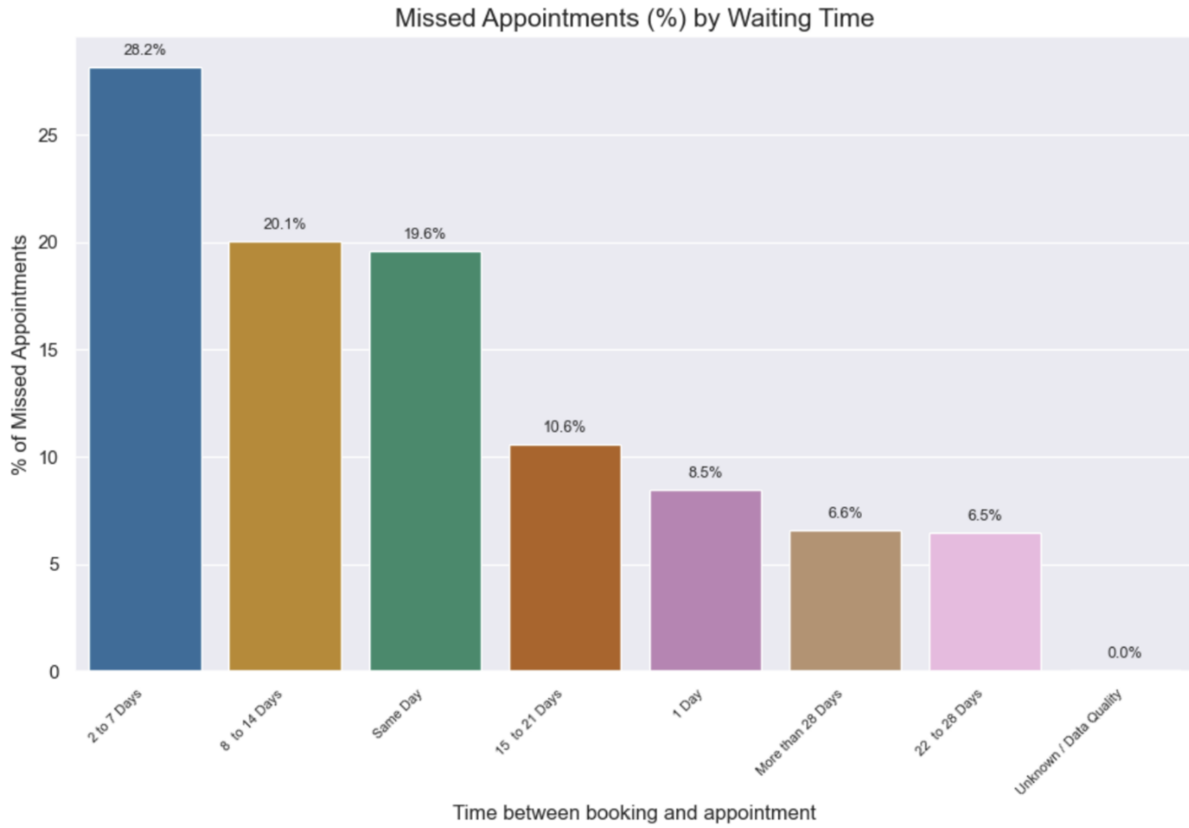
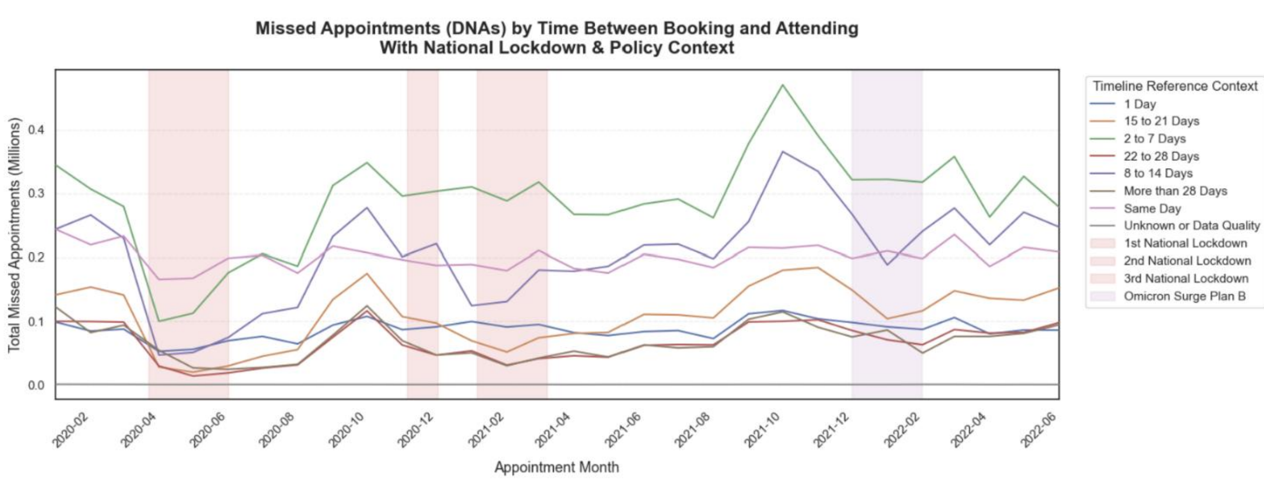


Figure 16: Missed Appointments by Wait time over the period Analysed



Time series of DNA appointments by wait time (January 2020 – June 2022):

- DNA appointments across all wait time categories dropped sharply at the March 2020 Covid lockdown, consistent with the overall reduction in booked appointments and potential recording failures during this period
- DNA levels recovered from mid-2020 and stabilised across most wait time categories throughout 2021
- The 2–7-day category remained the dominant source of DNA appointments throughout the period, with same-day DNA remaining consistently notable
- No single wait time category showed a material improvement trend over the period - missed appointments by wait time remained structurally stable rather than declining

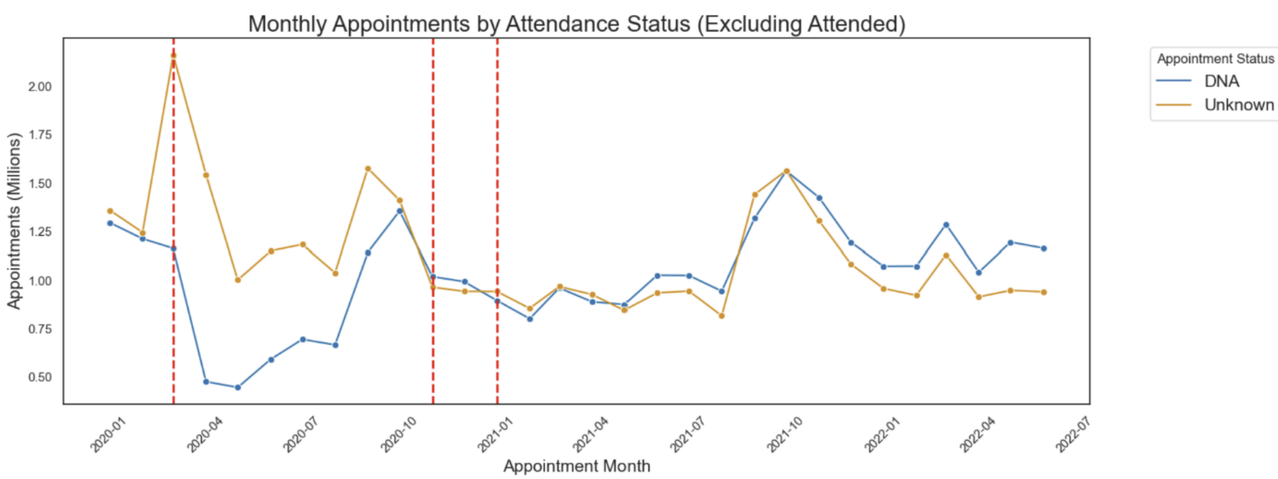
The structural stability of the DNA distribution over time is itself a significant finding. Despite the profound operational disruption of the Covid period, the distribution remained broadly consistent across the recovery period following lockdowns.

remained broadly consistent before and after lockdowns - the 2–7-day category retained its dominance and same-day remained notably high throughout. This suggests the DNA-by-wait-time pattern is not a consequence of the pandemic but a structural feature of NHS appointment behaviour that predates it and has persisted through it. Any intervention strategy aimed at reducing missed appointments therefore cannot rely on post-Covid normalisation to resolve the problem - it requires deliberate, targeted action by wait time category.

'Unknown' Status at the Covid Lockdown

Removing 'Attended' from the appointment status analysis (figure 17) revealed a critical finding: the 'Unknown' category showed a dramatic spike at the start of the first Covid lockdown in March 2020. Rather than representing genuine unknown attendance, this almost certainly reflects a systemic recording failure - the NHS data infrastructure was unable to capture appointment outcomes reliably during the disruption. The 'Unknown' category then remained elevated throughout the analysis period, suggesting the recording problem was never fully resolved.

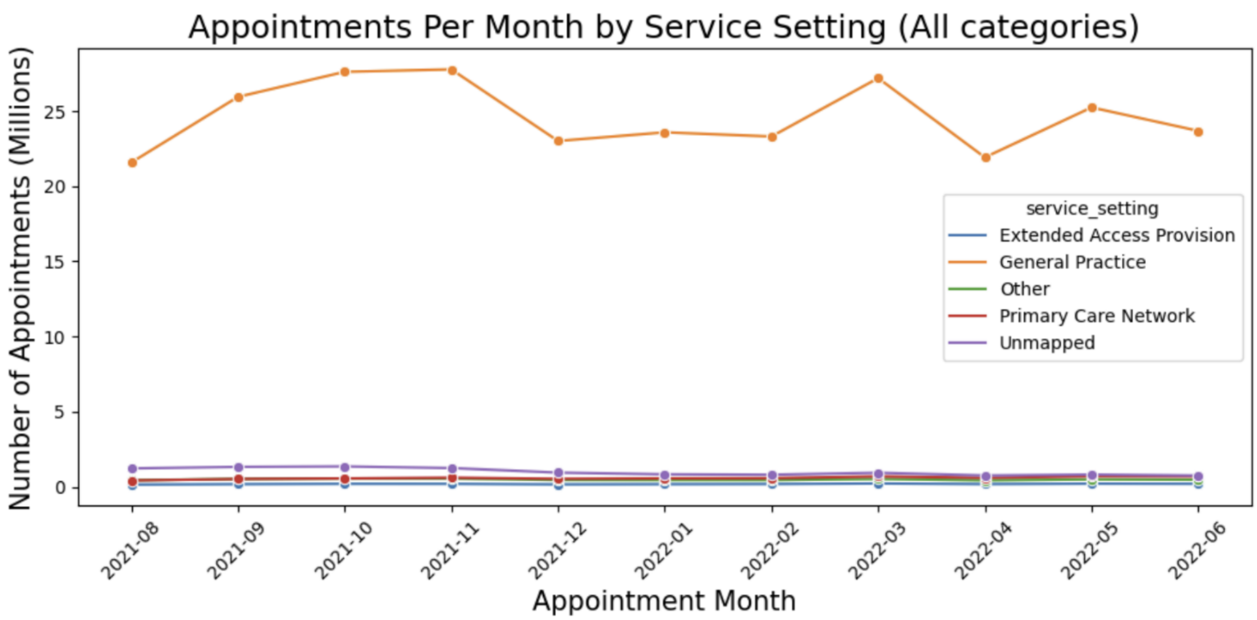
Figure 17: Missed Appointments by Wait time over the period Analysed (ex-Attended)



6. Service Setting and Context Type

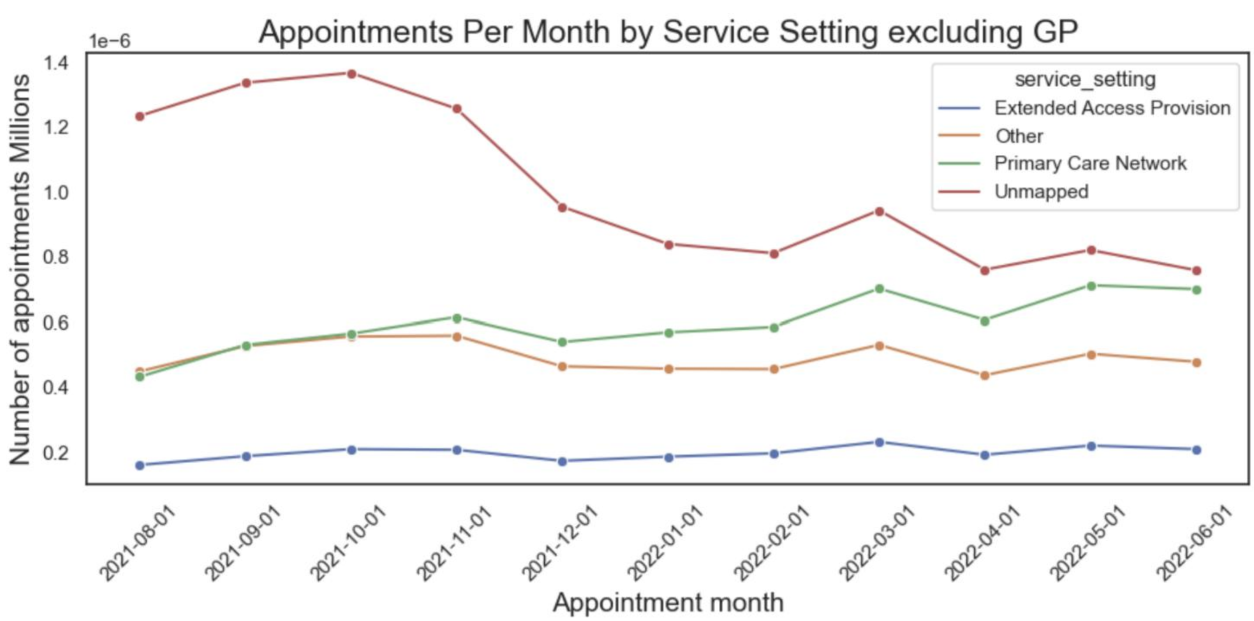
Figure 18 shows all service settings and the variation in appointment numbers over the time period. Of particular note is the dip from 12/2021 – 02/2022. GP appointments dominate making it difficult to gain any insights from the other service settings.

Figure 18: Appointments by Service Setting (All categories)



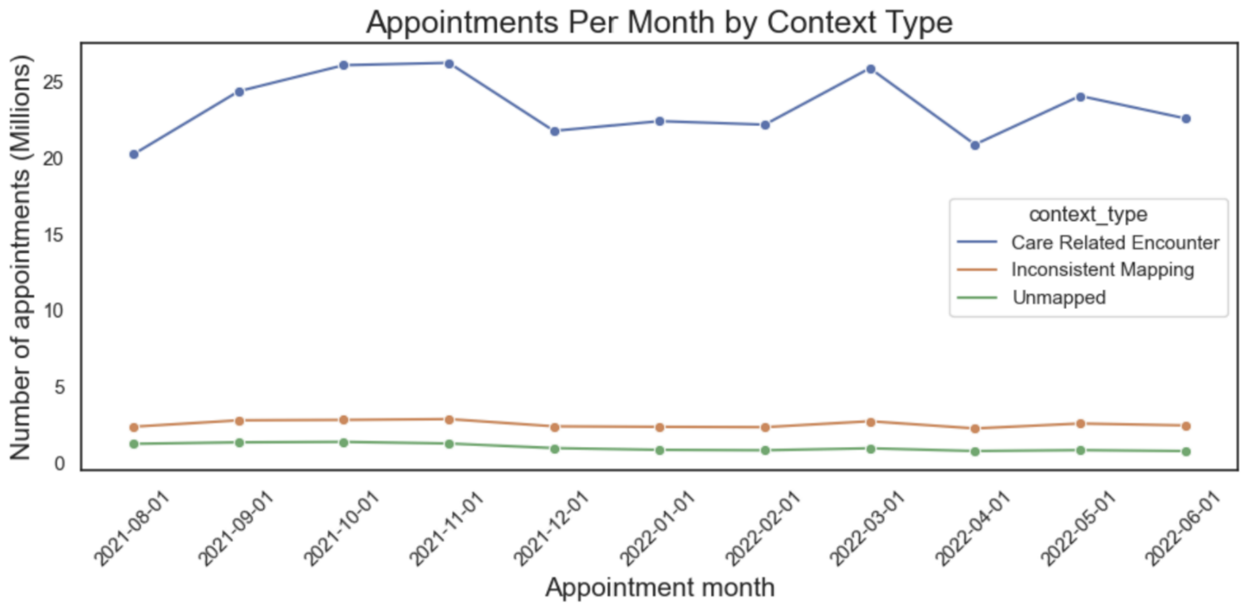
With GP removed from the analysis, the Unmapped category dominated all other service setting - accounting for over 250,000 appointments per week in the Autumn 2021 period. This is not a genuine service setting but a data quality failure: appointments that have not been correctly categorised at the point of recording.

Figure 19: Appointments by Service Setting (Ex GP)



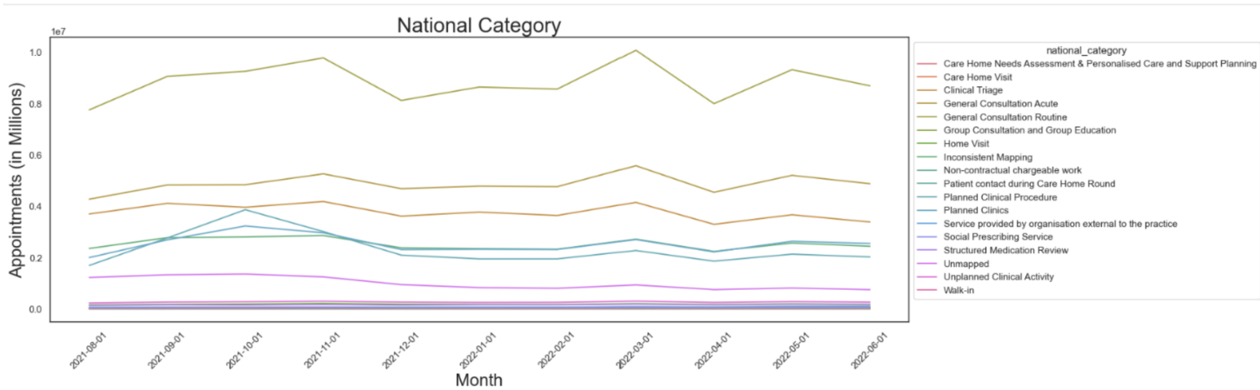
Context type analysis (figure 20) confirmed similar patterns: 'Inconsistent mapping' and 'Unmapped' were consistently present across the period, representing a systemic data capture problem. A notable pattern was observed in Autumn 2021: a significant drop in Unmapped appointments coincides with the introduction of the Learn from Patient Safety Events (LFPSE) service, a modernised digital recording system.

Figure 20: Appointments by Context Type



The national category breakdown (figure 21) showed Routine GP consultation as by far the largest category at 97.3 million appointments, followed by General Consultation Acute at 53.7 million. The lowest category was Group Consultation and Education at 60,632 - indicating the NHS has very limited capacity for group-based appointment models that could reduce per-appointment cost at scale.

Figure 21: Appointments by National Category

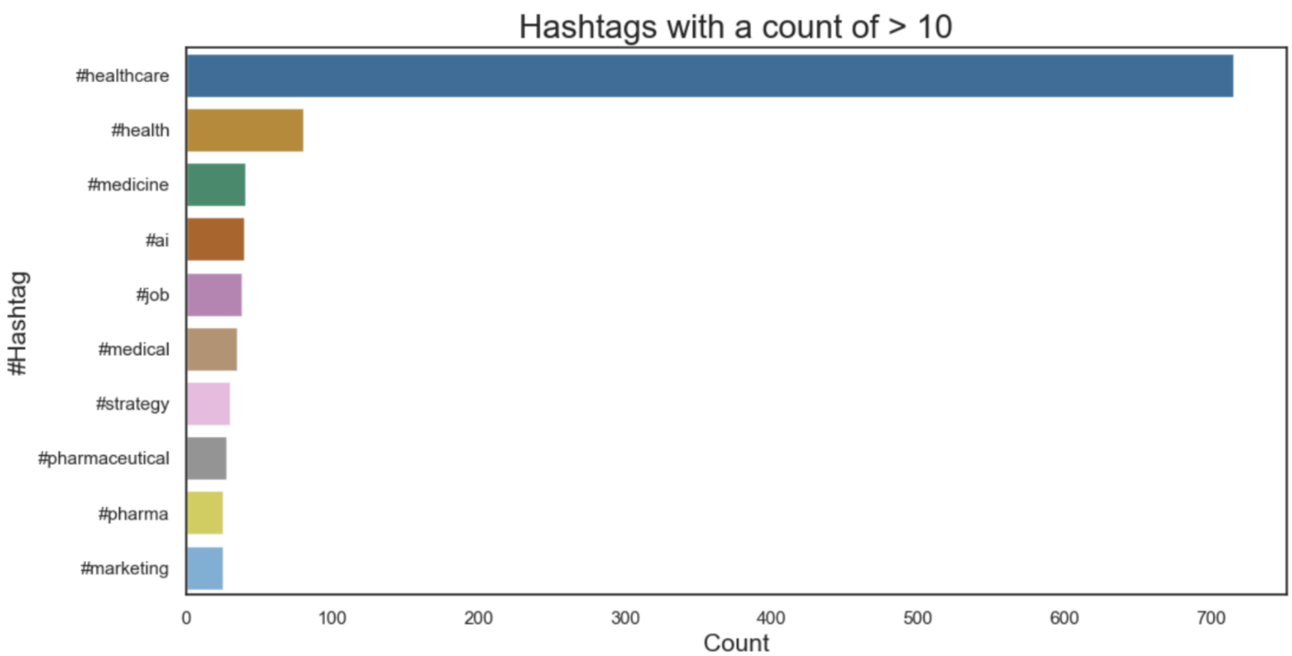


7. Twitter (X) / Social Media Data

The Twitter (X) dataset contained 1,174 records across 10 columns including hashtags, retweet counts, and favourite counts. #Healthcare was by far the most prominent hashtag - appearing nearly ten times more frequently than the second hashtag. Most tweets received zero retweets and zero likes. The dataset lacks the volume, geographic tagging, and metadata structure required for reliable sentiment analysis or regional segmentation.

This finding reinforces the recommendation to invest in structured social media data collection if social listening is a genuine strategic objective. The current dataset demonstrates the gap between ad hoc data and analytically useful data - a useful finding in itself, even if it cannot answer the original question.

Figure 22: Twitter (X) Analysis – Hashtags



Recommendations

1. Fix the Capacity Benchmark

- Replace the uniform 1.2 million daily figure with a demand-weighted model reflecting actual scheduled availability by day type (weekday, Saturday, Sunday, bank holiday). The current benchmark renders utilisation analysis unreliable.
- Model future demand using historical trend data - with explicit recognition that the 2020–2022 period is not typical due to Covid disruption, and the steep face-to-face appointment drop should not be treated as a stable baseline.
- Align capacity planning to seasonal patterns - prioritise resourcing during autumn and winter peaks. The data clearly shows predictable demand cycles that should inform staffing decisions proactively.

- Address Monday inflation: bank holiday displacement should be modelled explicitly rather than distorting the day-of-week capacity analysis.

2. Resolve the Data Quality Problem

- The 'Unknown / data quality' category must be addressed at source. It is the dominant value in multiple key metrics - including appointment duration, context type, and attendance status. Until it is resolved, no downstream analysis can be considered reliable.
- Standardise data collection and recording processes across all 106 locations. The regional variation in DNA and unknown rates suggests inconsistent recording practices, not just inconsistent patient behaviour.
- Clarify and document the capacity benchmark - what it represents, how it is derived, and under what conditions it should be updated. The current benchmark appears to be an arbitrary national figure not calibrated to actual scheduled capacity.

3. Target Missed Appointment Interventions by Risk Profile

- Same-day appointments have the highest DNA rate at 19.8% - explore whether these can be confirmed at time of booking or reminded closer to appointment time through automated systems.
- Deploy intervention resources differentially by region: NHS Black Country, NHS Greater Manchester, and Midlands/London clusters show the highest DNA rates and should be prioritised for targeted awareness campaigns and rescheduling infrastructure.
- Implement friction-free rescheduling - patient portals and apps that allow cancellation and rebooking without requiring a phone call are likely to convert DNA-bound patients into rescheduled appointments rather than lost slots.

4. Invest in Social Media Data Infrastructure

- Commission structured social media data collection with volume, geographic tagging, and consistent metadata if social listening is a genuine strategic objective - the current Twitter (X) dataset demonstrates what insufficient data looks like.
- Develop regular dashboards to monitor sentiment trends across major regions, drawing on multiple platforms rather than Twitter alone.

Limitations

Limitation	Impact	Mitigation Applied
Fixed daily capacity benchmark (1.2M)	Distorts utilisation rates - particularly on weekends. Weekday-only analysis developed as alternative.	Flagged throughout; weekday-adjusted analysis presented alongside benchmark figures
Unknown / data quality category	Dominant value in key metrics including duration, context type,	Identified as primary data constraint; findings scoped

	and attendance status - limits reliability across multiple analyses	accordingly and treated as directional
Unmapped appointment status (4–5%)	DNA and attended rates cannot be fully reconciled - true non-attendance may be higher than headline figures	Reported separately; clearly distinguished from confirmed DNA figures
Atypical time period (2020–2022)	Covid-19 disruption means trends are unlikely to represent normal operating conditions - particularly the face-to-face/telephone split	Findings treated as period-specific; Covid lockdown periods marked explicitly on time series charts
Monday displacement effect	Bank holiday rescheduling inflates Monday appointment counts, distorting day-of-week analysis	Flagged in day-of-week analysis; noted as structural not behavioural
Twitter (X) dataset quality	1,174 records insufficient for reliable sentiment analysis; no geographic tagging; sparse metadata	Reviewed for hashtag frequency patterns only; conclusions treated as illustrative rather than analytical
No demographic data	Cannot identify which patient groups drive missed appointments or why - a significant gap for targeted intervention design	Highlighted as key gap; demographic data collection recommended as first priority for future data development
Merged dataset limitations	Joining 'appointments_regional' and 'national_categories' on ICB code produces many-to-many matches - DNA-by-service-setting results treated as approximate	Directional conclusions only; precise attribution noted as limited

Further Analysis

The following extensions would materially improve the analytical value of this work:

- Demand-weighted capacity modelling - replace the fixed 1.2M benchmark with a dynamic model reflecting actual scheduled capacity by day type, location, and season. This is the single highest-impact near-term analytical extension.
- Demographic integration - incorporating patient demographic and socioeconomic data would enable identification of which groups are most likely to miss appointments, allowing evidence-based rather than generalised intervention design.
- Longitudinal trend analysis - extending the dataset beyond June 2022 would allow separation of Covid-era disruption from underlying structural trends, producing a reliable baseline for future capacity planning.
- Predictive DNA modelling - with demographic data added, a classification model could identify appointments at elevated risk of being missed, enabling proactive reminder targeting and resource reallocation.
- Multi-channel social media analysis - structured data collection across multiple platforms with consistent metadata would enable reliable sentiment analysis, regional segmentation, and public communication monitoring.

- Transport and contextual factor analysis - incorporating transport accessibility, weather, and employment patterns as external variables would improve understanding of structural drivers of missed appointments beyond patient behaviour alone.
- Group consultation capacity modelling - given that Group Consultation and Education accounted for only 60,632 appointments against 97.3 million routine GP consultations, modelling the potential impact of scaling group-based models warrants investigation as a cost-reduction strategy.

Repository Structure

Project files are available at: github.com/AndrewWillacy/Healthcare-and-Appointment-Utilisation-Analysis

File / Folder	Contents
actual_duration.csv	Appointment duration dataset
appointments_regional.csv	Regional appointments dataset (21,604 duplicates removed)
national_categories.csv	National categories dataset (converted from Excel)
NHS-ICB_names.csv	NHS ICB reference file downloaded from NHS website - merged with ‘appointments_regional’ for location name analysis
tweets.csv	Twitter/X dataset used for social media hashtag frequency analysis (converted from Excel)
sub_icb_boundaries_geojson.geojson	ONS Sub-ICB boundary file used for geographic DNA mapping by region
NHS_Healthcare_Jupyter_Notebook.ipynb	Full Python analysis notebook with all visualisations and commentary
Presentation_Slide_Deck.pptx	Slide deck summarising key findings and recommendations
NHS_Technical_Report.pdf	This document
metadata.txt	Field descriptions for all dataset columns